

- ALL
- 1
- 2
- 3
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- 6
- 7
- 8

2. Question 2

Consider an expression $a+1/a+\log [a] /a^2$ and a is a positive real integer. What can be minimum value of the expression?

Pick **ONE** option

- ☒ 2
- ☐ 3
- ☐ 1
- ☐ 0

Clear Selection

✓ Saved!

3. Question 3

ALL

4. Question 4

Louis College London offers six subjects: math, chemistry, physics, botany, zoology and anatomy. Each subject has its own teacher, and each teacher teaches a different number of sections from 1 - 6. How many sections does Dr. Noah teach?

1. Dr. Mason teaches 4 sections.
2. Anatomy has 5 sections.
3. Dr. Smith teaches 2 more sections than Dr. Noah. Neither teaches botany nor zoology.
4. Dr. Thomas teaches less than 6 sections of physics.
5. There are more sections of chemistry than math. Dr. Lucas does not teach chemistry.
6. There are 3 more sections of zoology than botany.

Pick **ONE** option

☐ zoology

☐ chemistry

☐ physics

☐ mathematics

Clear Selection

- ALL
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

5. Question 5

"CURATED" is encoded as "UFEASVD". How would "AMAZING" be encoded?

Pick **ONE** option

- ☐ JOHZBNB
- ☐ HOJZBNB
- ☐ BNBZHOJ
- ☐ BNBJOH

Clear Selection

- ALL
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

6. Question 6

A man spent 1/3rd of his life as a bachelor. After ten years of his marriage a son was born to him. The father and son went on a vacation together at the time the father's age was double that of his son. What was the father's age at the time of the vacation?

Pick **ONE** option

- ☐ 50
- ☐ 60
- ☐ 70
- ☐ 80

Clear Selection

7. Question 7

ALL



3

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10

11

Clear Selection

7. Question 7

Let $X_1, X_2, \dots, X_{75}$ be *i.i.d.*, each with expected value $\mu = E(X_i) = 2$, and variance $\sigma^2 = \text{Var}(X_i) = 3$. Approximate $P(X_1 + X_2 + \dots + X_{75} > 120)$ using the central limit theorem. For the solution, it can be assumed that $P(|Z| < 2) \approx 0.95$ where Z is a random variable that has a standard normal distribution.

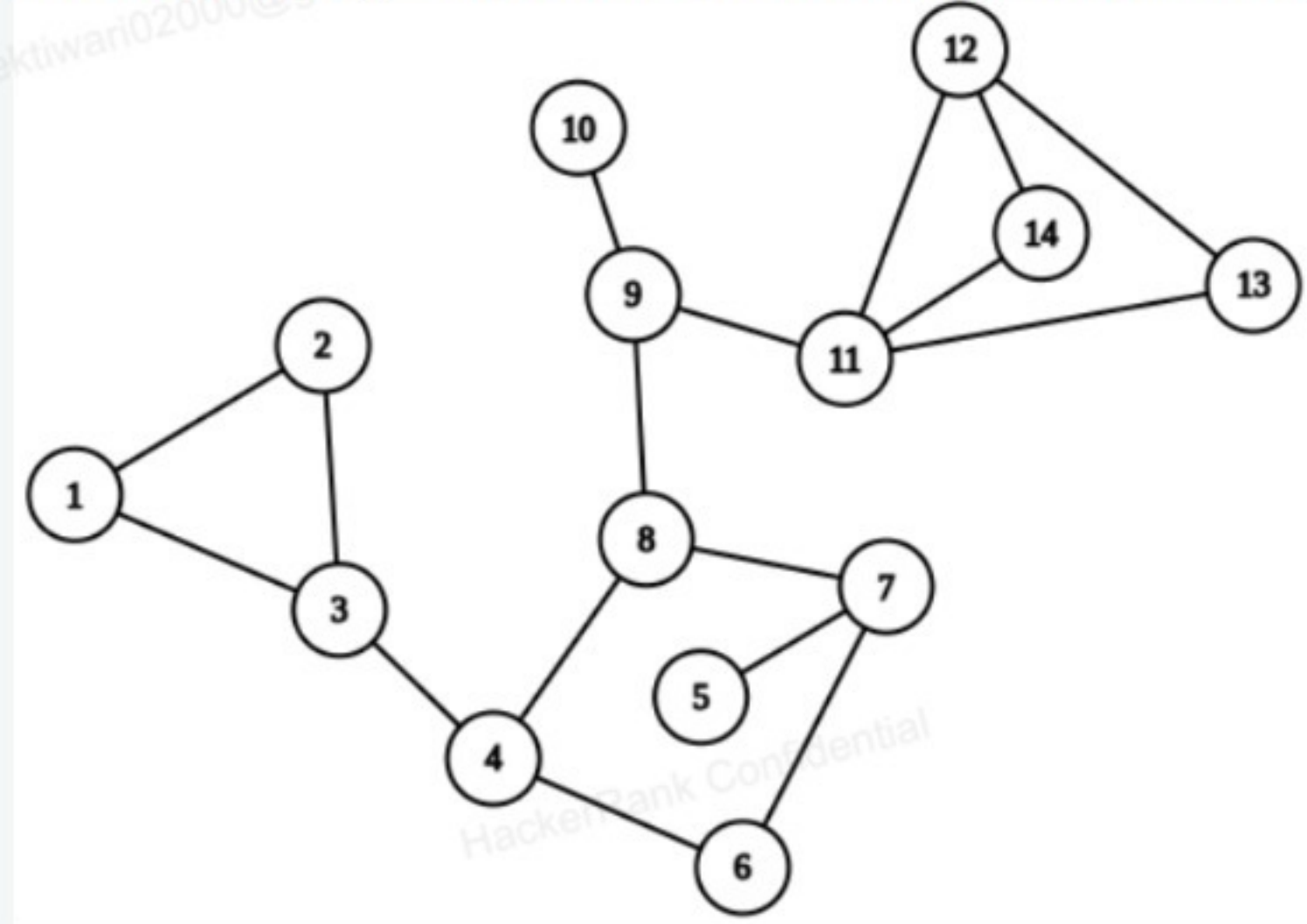
Pick **ONE** option

☐ 0.95

☐ 0.85☐ 0.975☐ 0.6

Clear Selection

Which of the following is/are a valid DFS traversal starting from node 9 of the given graph? There may be more than 1.



Pick **ONE OR MORE** options

- ☐ 9, 10, 11, 13, 12, 14, 8, 7, 5, 6, 4, 3, 2, 1
- ☐ 9, 10, 11, 13, 14, 12, 8, 7, 5, 6, 4, 3, 2, 1
- ☐ 9, 8, 4, 3, 1, 2, 6, 7, 5, 10, 11, 14, 13, 12
- ☐ 9, 8, 4, 3, 1, 2, 7, 6, 5, 10, 11, 14, 13, 12

1h 15m left

ALL

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18

18. Question 18

Given a list of strings comprised of a name and a Roman numeral, sort the list first by name, then by the decimal value of the Roman numeral.

In Roman numerals, a value is not repeated more than three times. At that point, a smaller value precedes a larger value to indicate subtraction. For example, the letter I represents the number 1, and V represents 5. Reason through the formation of 1 to 10 below, and see how it is applied in the following lines.

- I, II, III, IV, V, VI, VII, VIII, IX, and X represent 1 through 10.
- XX, XXX, XL, and L are 20, 30, 40, and 50.
- For any other two-digit number < 50, concatenate the Roman numeral(s) that represent its multiples of ten with the Roman numeral(s) for its values < 10. For example, 43 is 40 + 3 = 'XL' + 'III' = 'XLIII'

Example

names = ['Steven XL', 'Steven XVI', 'David IX', 'Mary XV', 'Mary XIII', 'Mary XX']

The result with Roman numerals is the expected return value. Written in decimal and sorted, they

Language Python 3

Environment

Autocomplete Ready

```
1 > #!/bin/python3 ...
10
11 #
12 # Complete the 'sortRoman' function below.
13 #
14 # The function is expected to return a STRING_ARRAY.
15 # The function accepts STRING_ARRAY names as parameter.
16 #
17
18 def sortRoman(names):
19     # Write your code here
20
21 > if __name__ == '__main__': ...
```

Test Results

Custom Input

Run Code

Run Tests

Submit

Line: 10 Col: 1

1h 14m left

ALL

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18

The result with Roman numerals is the expected return value. Written in decimal and sorted, they are ['David 9', 'Mary 13', 'Mary 15', 'Mary 20', 'Steven 16', 'Steven 40']. The return array is ['David IX', 'Mary XIII', 'Mary XV', 'Mary XX', 'Steven XVI', 'Steven XL'].

Function Description

Complete the function *sortRoman* in the editor below.

sortRoman has the following parameter:

- names[n]*: an array of strings comprised of names and roman numerals

Returns:

- string[n]*: an array of strings sorted first by given name, then by ordinal

Constraints

- $1 \leq n \leq 50$
- Each *names[i]* is a single string composed of 2 space-separated values: *givenName* and *romanNumeral*.
- romanNumeral* represents a number between 1 and 50, inclusive.
- $1 \leq |givenName| \leq 20$
- Each *givenName* starts with an uppercase letter *ascii*[A-Z] which is followed by lowercase letters *ascii*[a-z].
- There is a space between *givenName* and *romanNumeral*
- Each *names[i]* is distinct.

Language Python 3 Environment Autocomplete Ready

```
1 > #!/bin/python3 ...
10
11 #
12 # Complete the 'sortRoman' function below.
13 #
14 # The function is expected to return a STRING_ARRAY.
15 # The function accepts STRING_ARRAY names as parameter.
16 #
17
18 def sortRoman(names):
19     # Write your code here
20
21 > if __name__ == '__main__': ...
```

Test Results Custom Input

Run Code Run Tests Submit

Line: 10 Col: 1

1h 14m left

ALL

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Sample Input

```
STDIN      Function
-----
2  →  names[] size n = 2
Louis IX  →  names = ['Louis IX',
'Louis VIII']
Louis VIII
```

Sample Output

```
Louis VIII
Louis IX
```

Explanation

Sort first by *givenName* then, if *givenName* is not unique, by the value of the Roman numeral. In decimal, the list is sorted *['Louis 8', 'Louis 9']*.

▼ Sample Case 1

Sample Input

```
STDIN      Function
-----
2  →  names[] size n = 2
Philippe I  →  names = ['Philippe I',
'Philip II']
Philip II
```

Sample Output

```
Philip II
Philippe I
```

Language Python 3 Environment

Autocomplete Ready

```
1 > #!/bin/python3...
10
11 #
12 # Complete the 'sortRoman' function below.
13 #
14 # The function is expected to return a STRING_ARRAY.
15 # The function accepts STRING_ARRAY names as parameter.
16 #
17
18 def sortRoman(names):
19     # Write your code here
20
21 > if __name__ == '__main__':...
```

Test Results Custom Input

Run Code Run Tests Submit

Line: 10 Col: 1

1h 12m left

ALL

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17. Question 17

A software developer is working on a real-time messaging platform used by millions of users. The platform needs to handle a high volume of messages efficiently while minimizing latency.

The developer has n messages to deliver. $platformTime[i]$ is the time taken to process and deliver the i^{th} message, $supportTime[i]$ is the time taken for a supporting service (e.g., caching service, message queue) to process the i^{th} message, where $0 \leq i < n$. The supporting service and platform work in parallel.

The task is to minimize the time to deliver all the messages through strategic use of the platform's resources and supporting services. Return this minimum time value.

Note: All supporting services work in parallel to process different messages simultaneously.

Example

$n = 6$
 $platformTime = [9, 6, 8, 6, 4, 4]$
 $supportTime = [4, 3, 7, 7, 9, 8]$

The messages from indices 0 through 3 can be processed and delivered by the supporting

Language Python 3 Environment

```
1 > #!/bin/python3...
10
11 #
12 # Complete the 'getMinimumTime' function below.
13 #
14 # The function is expected to return an INTEGER.
15 # The function accepts following parameters:
16 # 1. INTEGER_ARRAY platformTime
17 # 2. INTEGER_ARRAY supportTime
18 #
19
20 def getMinimumTime(platformTime, supportTime):
21     # Write your code here
22
23 > if __name__ == '__main__':...
```

Test Results Custom Input

Run Code Run Tests Submit

1h 12m left

ALL

Example

$n = 6$
 $platformTime = [9, 6, 8, 6, 4, 4]$
 $supportTime = [4, 3, 7, 7, 9, 8]$

The messages from indices 0 through 3 can be processed and delivered by the supporting services, while the platform's resources handle messages at indices 4 and 5. The supporting services will complete the task in a maximum of $\max(\{4, 3, 7, 7\}) = 7$ units of time, and the platform processes it in $4 + 4 = 8$ units of time. Hence, the total time to deliver all the messages is $\max(7, 8) = 8$ time units.

Function Description

Complete the function `getMinimumTime` in the editor below.

$getMinimumTime$ takes the following parameter(s):

$int\ platformTime[n]$: the time taken by the platform's resources to process and deliver each message

$int\ supportTime[n]$: the time taken by the supporting services to process and deliver each message

Returns

int : the minimum total time to deliver all the messages

Constraints

Language Python 3

Environment

Autocomplete Ready

1 > `#!/bin/python3...`

10

11 #

12 # Complete the 'getMinimumTime' function below.

13 #

14 # The function is expected to return an INTEGER.

15 # The function accepts following parameters:

16 # 1. INTEGER_ARRAY platformTime

17 # 2. INTEGER_ARRAY supportTime

18 #

19

20 `def getMinimumTime(platformTime, supportTime):`

21 `# Write your code here`

22

23 > `if __name__ == '__main__':...`

Test Results

Custom Input

Run Code

Run Tests

Submit

Line: 10 Col: 1

Type here to search

Sailing

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1h 12m left

int: the minimum total time to deliver all the messages

Constraints

- $1 \leq n \leq 2 * 10^5$
- $1 \leq platformTime[i], supportTime[i] \leq 10^9$

Input Format For Custom Testing

Sample Case 0

Sample Input For Custom Testing

STDIN	FUNCTION
6	platformTime[] size n =
6	
2	platformTime = [2, 1,
3, 4, 5, 6]	
1	
3	
4	
5	
6	
6	supportTime[] size n =
6	
5	supportTime = [5, 2, 4,
6, 1, 4]	
2	
4	
6	
1	
4	

Language Python 3 Environment Autocomplete Ready

```
1 > #!/bin/python3...
10
11 #
12 # Complete the 'getMinimumTime' function below.
13 #
14 # The function is expected to return an INTEGER.
15 # The function accepts following parameters:
16 # 1. INTEGER_ARRAY platformTime
17 # 2. INTEGER_ARRAY supportTime
18 #
19
20 def getMinimumTime(platformTime, supportTime):
21     # Write your code here
22
23 > if __name__ == '__main__':...
```

Line: 10 Col: 1

Test Results Custom Input Run Code Run Tests Submit

1h 10m left

ALL

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18

Explanation
The platform processes the 2nd and 4th messages itself and assigns the rest to supporting services. The platform processes the messages in $1 + 4 = 5$ units of time, while the other services complete the task in $\max(\{5, 4, 1, 4\}) = 5$ units of time.

▼ Sample Case 1

Sample Input For Custom Testing

STDIN	FUNCTION
7	platformTime[] size n =
7	
9	platformTime = [9, 4,
9, 1, 6, 3, 8]	
4	
9	
1	
6	
3	
8	
7	supportTime[] size n =
7	
5	supportTime = [5, 4, 1,
9, 1, 1, 6]	
4	
1	
9	
1	
1	
6	

Sample Output

Language Python 3 Environment Autocomplete Ready

```
1 > #!/bin/python3 ...
10
11 #
12 # Complete the 'getMinimumTime' function below.
13 #
14 # The function is expected to return an INTEGER.
15 # The function accepts following parameters:
16 # 1. INTEGER_ARRAY platformTime
17 # 2. INTEGER_ARRAY supportTime
18 #
19
20 def getMinimumTime(platformTime, supportTime):
21     # Write your code here
22
23 > if __name__ == '__main__':...
```

1h 10m left

ALL

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▼ Sample Case 1

Sample Input For Custom Testing

STDIN	FUNCTION
7 7 9 9, 1, 6, 3, 8] 4 9 1 6 3 8	platformTime[] size n =
7 7 5 9, 1, 1, 6] 4 1 9 1 1 6	supportTime[] size n =

Sample Output

6

Explanation

The platform processes messages at indices 3 and 5 in $1 + 3 = 4$ units. It assigns the rest to the supporting services. Other services complete the task in $\max(\{5, 4, 1, 1, 6\}) = 6$ units of time.

Language Python 3 Environment Autocomplete Ready

```
1 > #!/bin/python3 ...
10
11 #
12 # Complete the 'getMinimumTime' function below.
13 #
14 # The function is expected to return an INTEGER.
15 # The function accepts following parameters:
16 # 1. INTEGER_ARRAY platformTime
17 # 2. INTEGER_ARRAY supportTime
18 #
19
20 def getMinimumTime(platformTime, supportTime):
21     # Write your code here
22
23 > if __name__ == '__main__': ...
```

Test Results Custom Input

Run Code Run Tests Submit

Line: 10 Col: 1



ALL



1

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1. Question 1

In 1990, about 50% of Americans considered the low quality of education as a serious problem for the country. What is the chance that the sample proportion will reflect the proportion of the total population with an accuracy of 3 percent for a random poll of 1,500 Americans?

Quantile table for standard normal distribution:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936

	z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990	
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993	
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995	
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997	
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	

- ALL
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8

2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Pick ONE option

☐ 0.999

☐ 0.98

☐ 0.9

☐ 0.8

Clear Selection

- ALL
- 1
- ✓
- 3
- 4
- 5
- 6
- 7
- 8

3. Question 3

Three people A, B and C are standing in queue. There are five people from A through B and ten people from B through C. It is given that the extremes of the queue can be occupied by A, B and C only. Which of the following can be the number of people standing in the queue?

Pick **ONE** option

- ☒ 10
- ☐ 15
- ☐ 20
- ☐ Both A and B

Clear Selection

4. Question 4